

A. Training Phase: Stochastic Sampling & Joint Dual Objective

Visual Pipeline (Frozen ViT → HEDO → SSD)

Latent $\mathbf{z}_I = \boldsymbol{\mu}_I + \boldsymbol{\varepsilon}_I \odot \boldsymbol{\sigma}_I \rightarrow \text{Embedding } \mathbf{e}_I$

Textual Pipeline (Frozen RoBERTa → HEDO → SSD)

Latent $\mathbf{z}_T = \boldsymbol{\mu}_T + \boldsymbol{\varepsilon}_T \odot \boldsymbol{\sigma}_T \rightarrow \text{Embedding } \mathbf{e}_T$

Joint Loss: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{InfoNCE}}(\mathbf{e}_I, \mathbf{e}_T; \tau) + 0.01 \cdot \mathcal{D}_{\text{SKL}}(p_I \parallel p_T)$

Multi-Positive Contrastive Alignment (5:1) + Symmetric FP32 KL Regularization

B. Test-Time Retrieval: Deterministic Unimodal Evaluation (Zero Cross-Modal Leakage)

Query Image $\mathbf{I} \in \mathbb{R}^{3 \times 224 \times 224}$

Deterministic Mean: $\hat{\mathbf{e}}_I = \text{Norm}(\mathbf{h}_{\text{pool}, I} + \alpha \mathbf{W}_{\text{out}} \boldsymbol{\mu}_I)$

Gallery Text Captions $\mathcal{T}_{\text{gallery}}$

Deterministic Mean: $\hat{\mathbf{e}}_T = \text{Norm}(\mathbf{h}_{\text{pool}, T} + \alpha \mathbf{W}_{\text{out}} \boldsymbol{\mu}_T)$

Cosine Similarity Matrix $\mathbf{S} = \hat{\mathbf{e}}_I \hat{\mathbf{e}}_T^\top \rightarrow \text{Rank @ K (1, 5, 10) \& MedR}$

No Cross-Modal Attention | No Inter-Modality Recurrent Communication